

IUOE APTITUDE TEST - 10 MINUTE CRAM SHEET

All formulas, conversions, and exam traps on one page

MATH FORMULAS

Percent Grade = $(\text{Rise} / \text{Run}) \times 100$

Bank Cubic Yards = Loose Cubic Yards / Swell Factor

Cut/Fill Volume = Length x Width x (Cut or Fill Depth)

Equipment Production = Volume / Time

Fuel Consumption = Hours / Fuel Tank Capacity

HYDRAULIC PRESSURE

1 PSI = 6.89 kPa

1 Bar = 14.5 PSI

Flow Rate = Pump Displacement x RPM

COMMON UNIT CONVERSIONS

1 yard = 3 feet

1 cubic yard = 27 cubic feet

1 mile = 5,280 feet

1 gallon = 3.785 liters

1 ton = 2,000 pounds

EXAM TRAPS TO AVOID

- Watch unit conversions (cubic yards vs feet)
- Percent grade is RISE/RUN, not vice versa
- Mechanical questions test your reasoning, not memory
- Time management: move on if stuck for 2+ minutes
- Read questions twice - look for "NOT" or "EXCEPT"

EQUIPMENT SPECS TO KNOW

- Excavator bucket sizes: 0.5 to 3+ cubic yards
- Dozer blade capacities: 5 to 15 cubic yards
- Loader bucket: 2 to 5 cubic yards
- Grader: 14 to 20 foot blade width

Good luck. You already know this. Just trust yourself.

Built by operators. for operators.